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(54) SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF

(71) Applicant: **Winbond Electronics Corp.**, Taichung City (TW)

(72) Inventors: **Li-Min Yan**, Tainan (TW); **Pin-Hung Chen**, Tainan (TW); **Ji-Min Lin**, Tainan (TW)

(73) Assignee: **Winbond Electronics Corp.**, Taichung City (TW)

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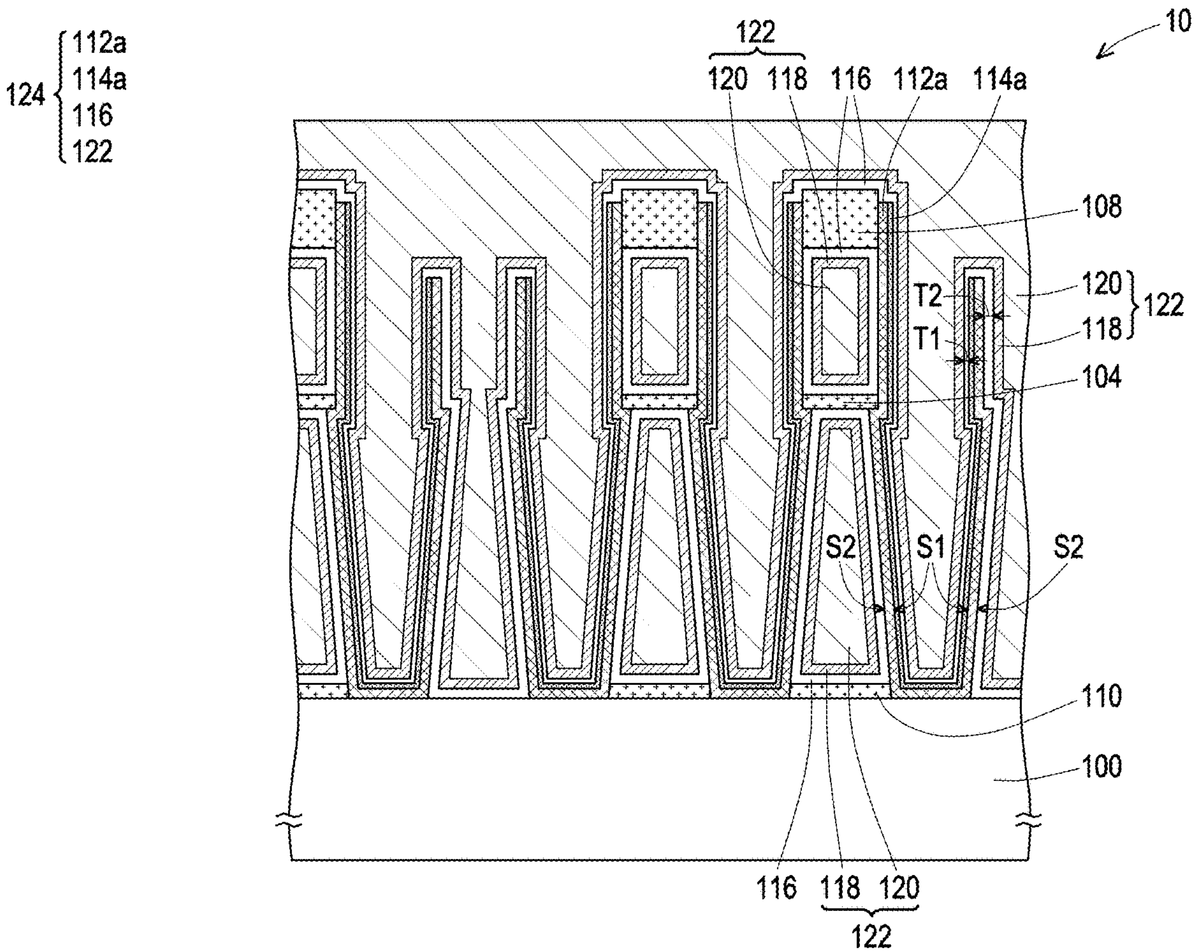
(57) **ABSTRACT**

A semiconductor structure including a substrate and a capacitor is provided. The capacitor is located on the substrate. The capacitor includes a first electrode, a second electrode, a first dielectric layer, and a second dielectric layer. The first electrode is located on the substrate. The first electrode has a first surface and a second surface opposite to each other. The second electrode is located on the first electrode. The first dielectric layer is located between the first surface and the second electrode. The second dielectric layer is located between the first dielectric layer and the second electrode and between the second surface and the second electrode.



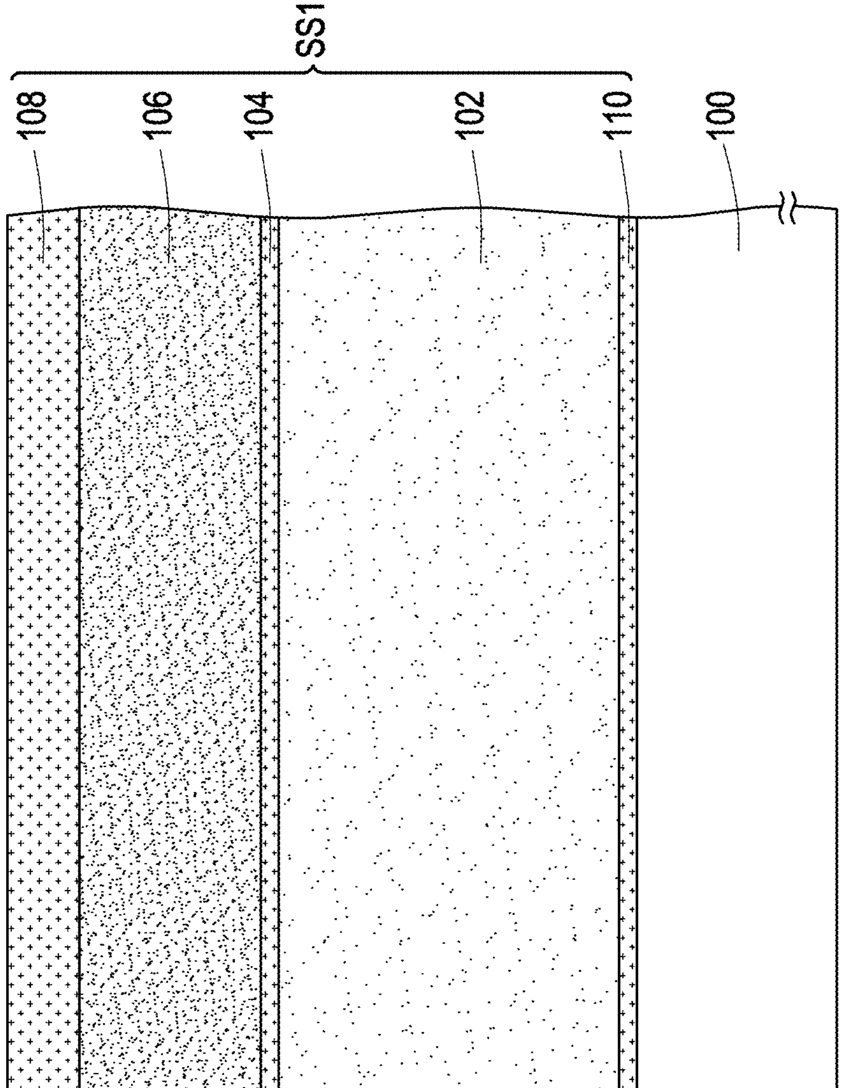


FIG. 1A

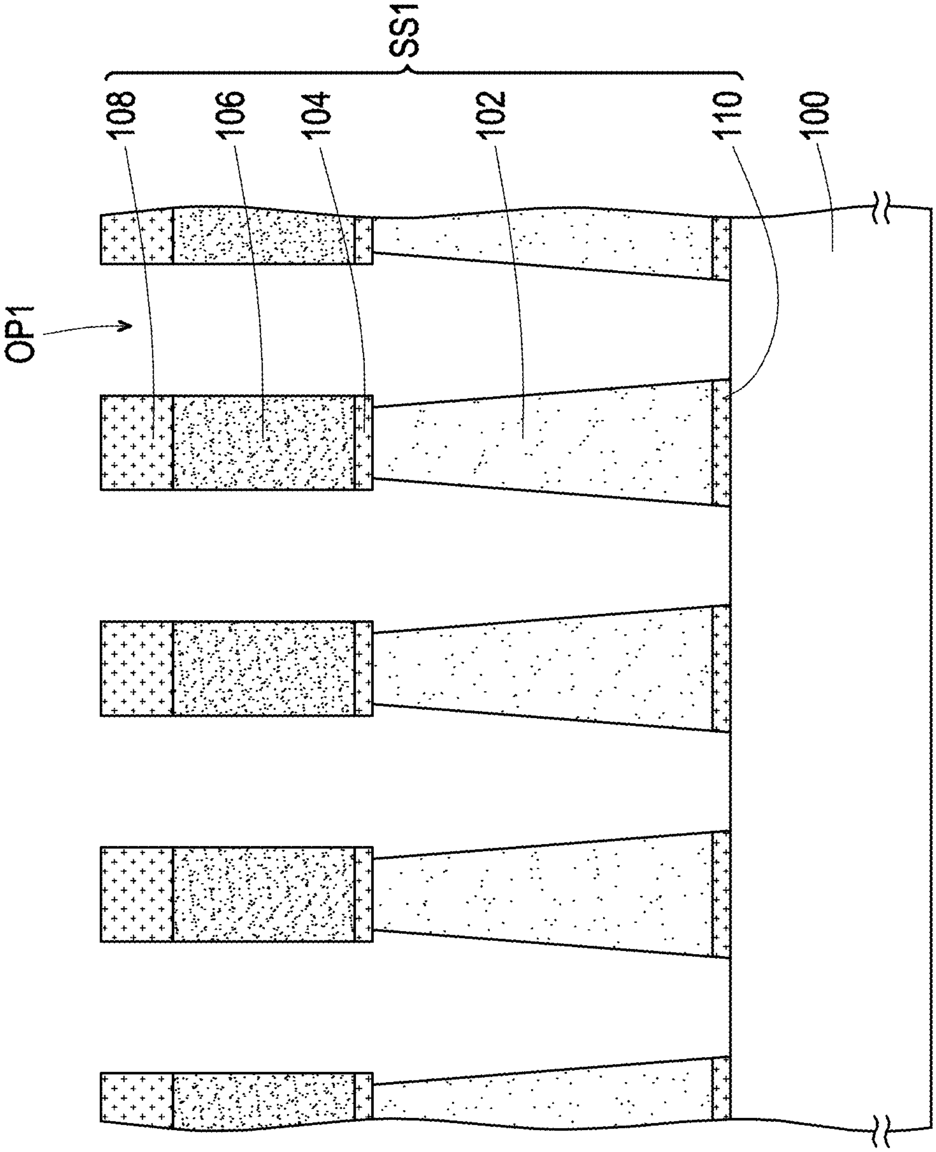


FIG. 1B

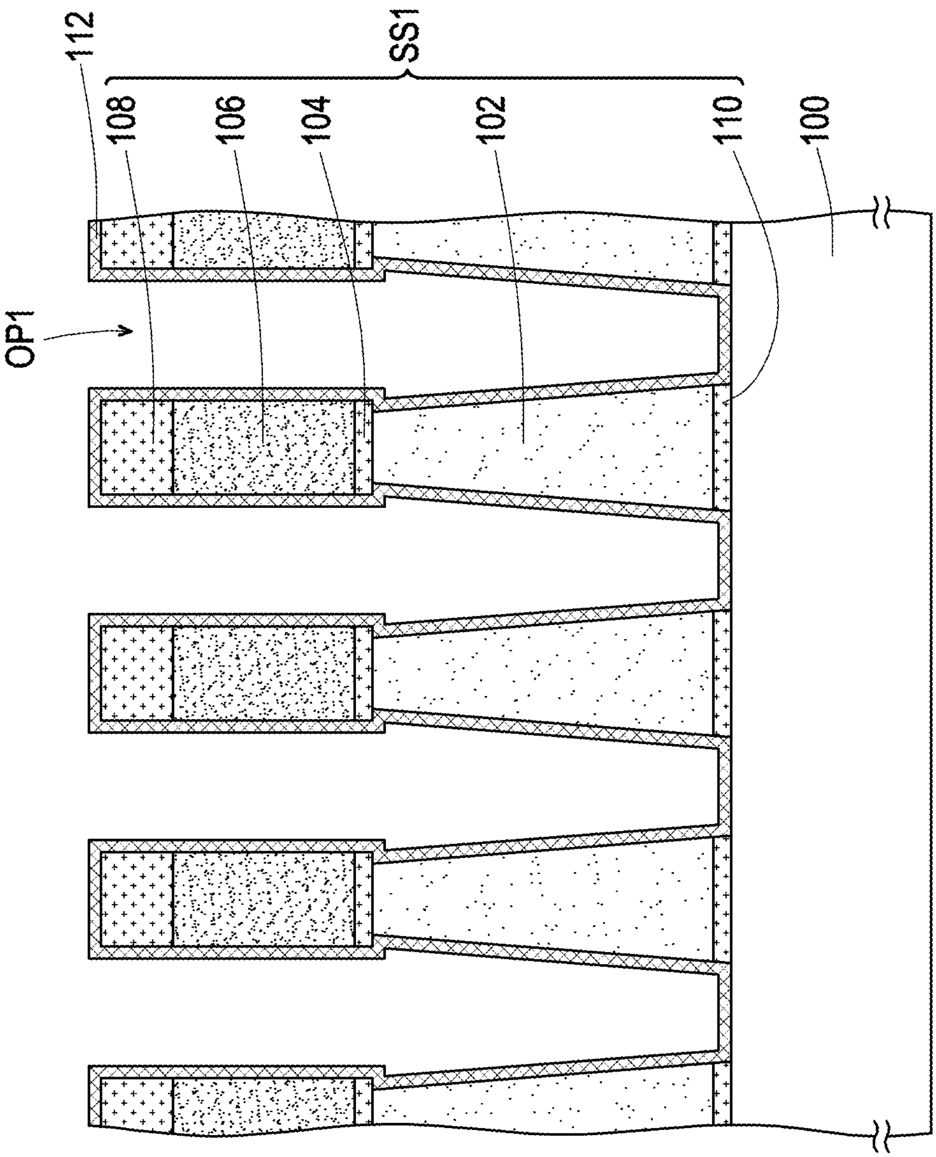


FIG. 1C

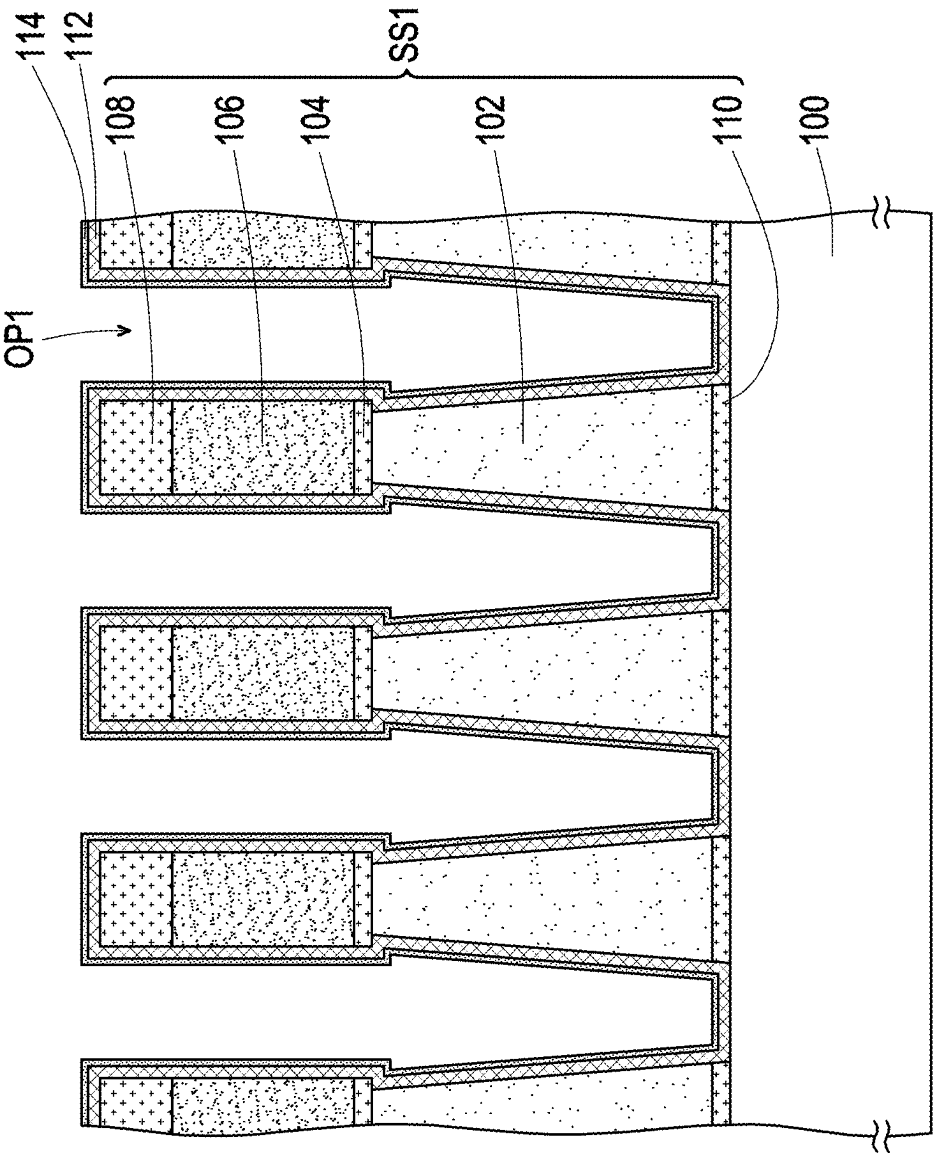


FIG. 1D

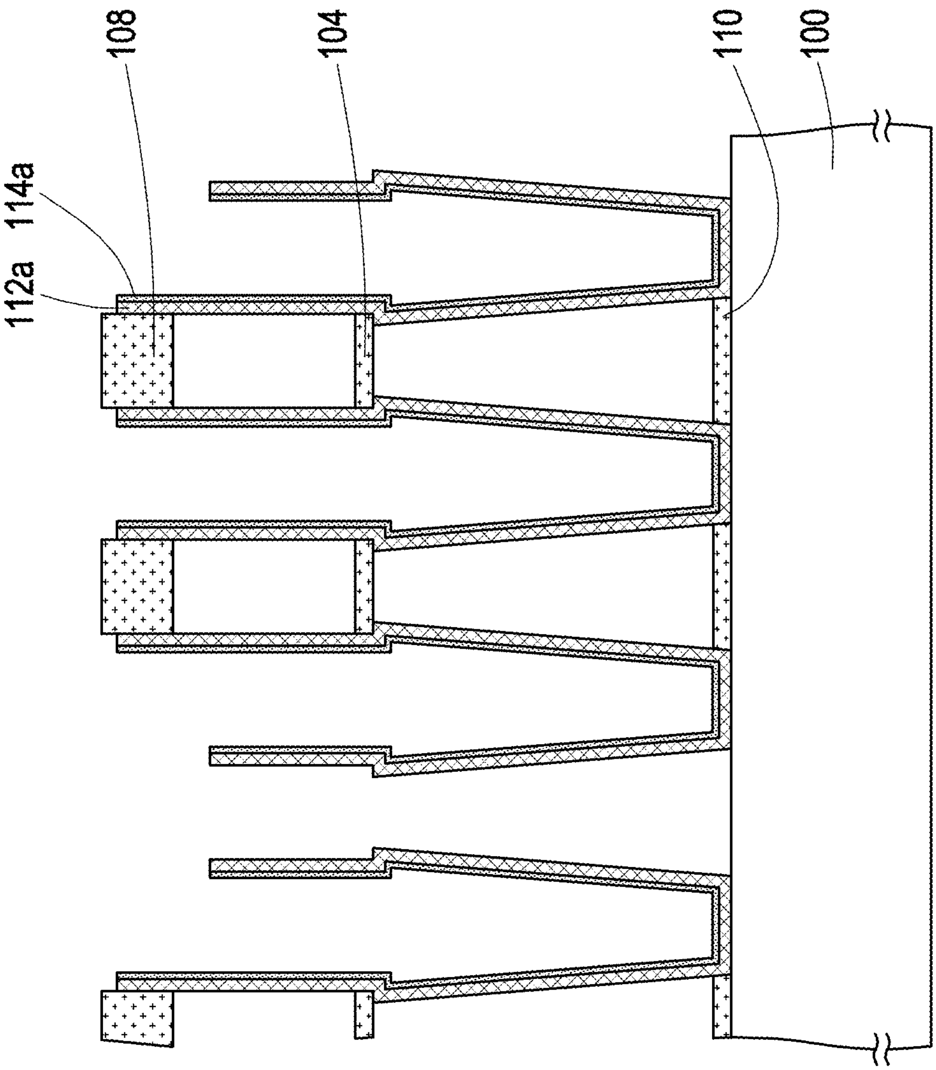


FIG. 1E

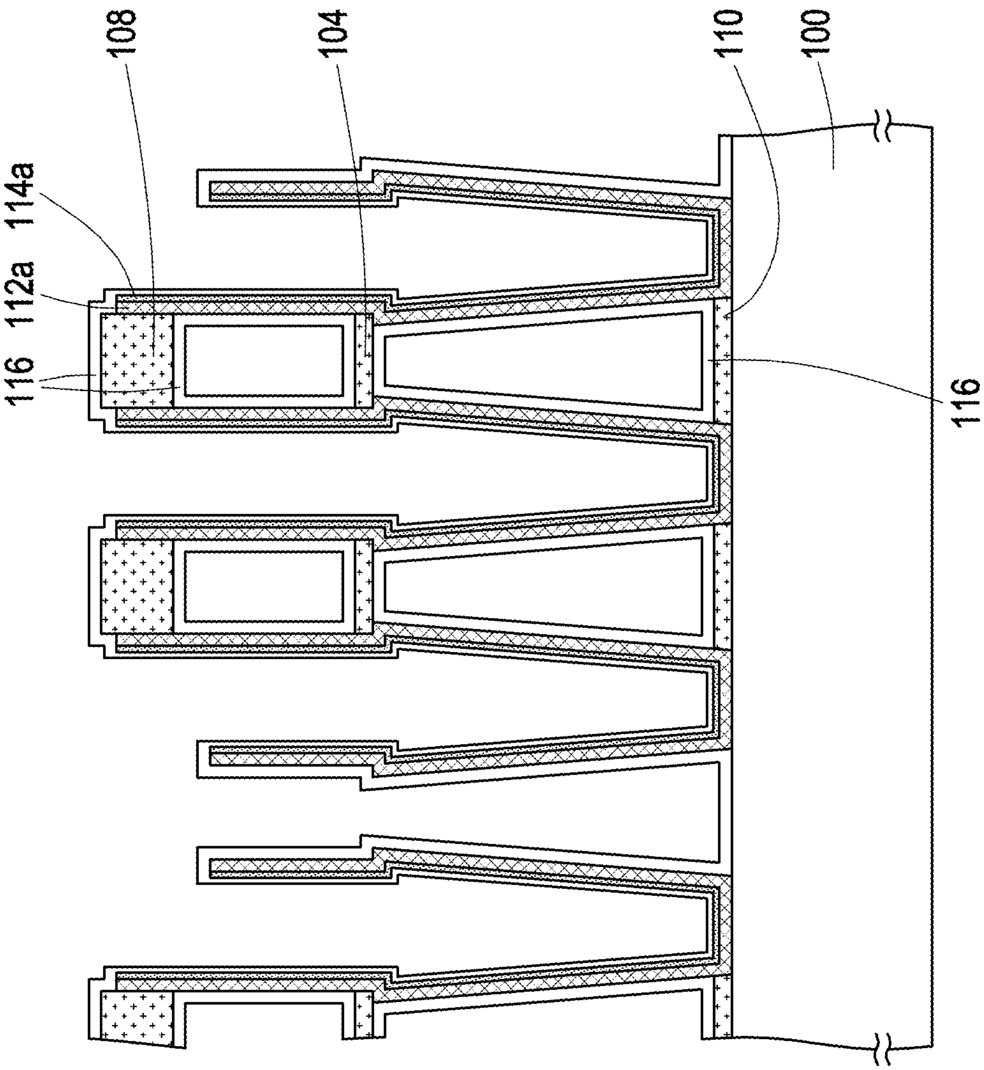


FIG. 1F

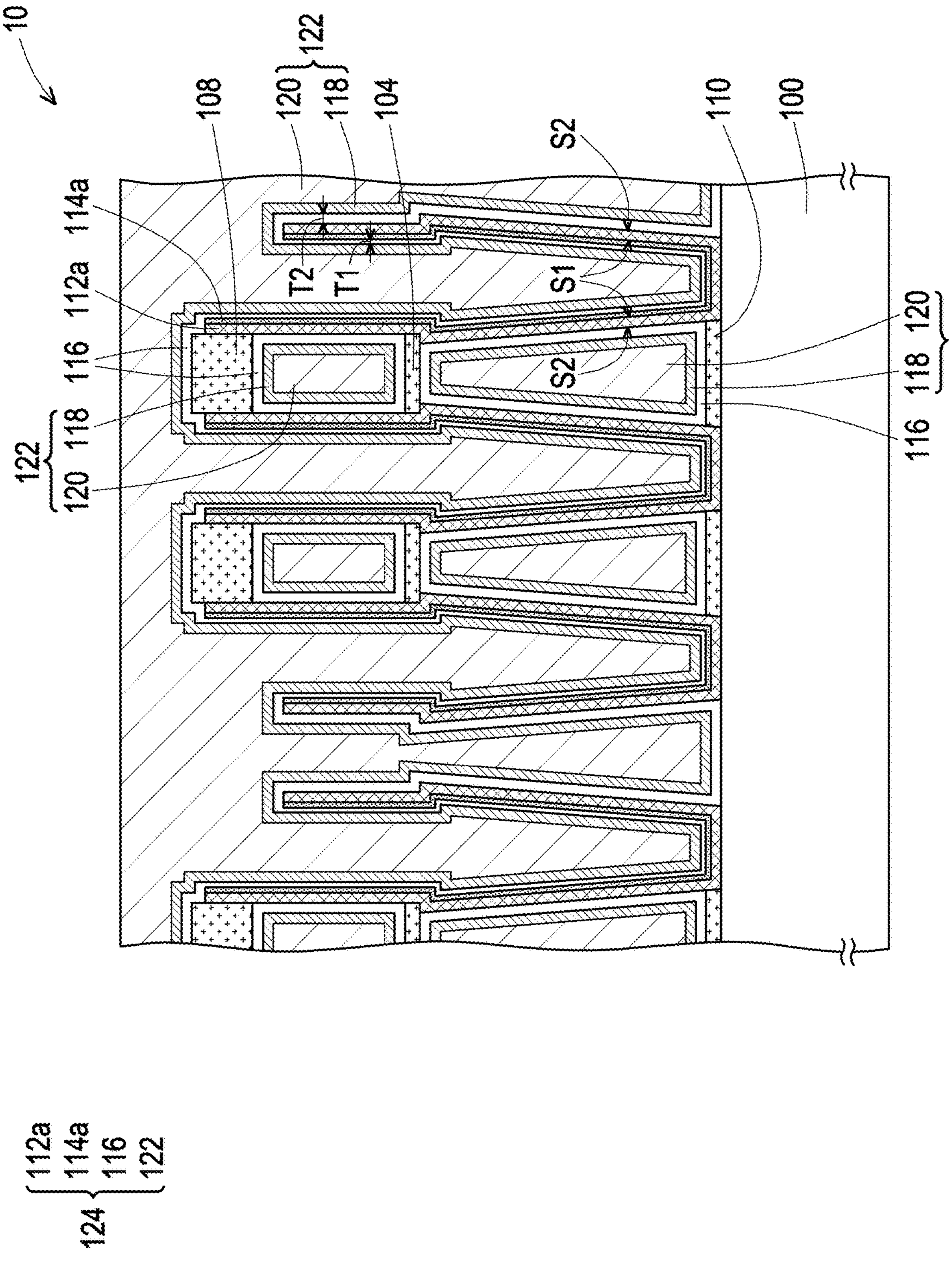

10
122
120
118
108
104
114a
112a
116
110
100
S2
S1
T1
T2
124
112a
114a
116
122

FIG. 1G

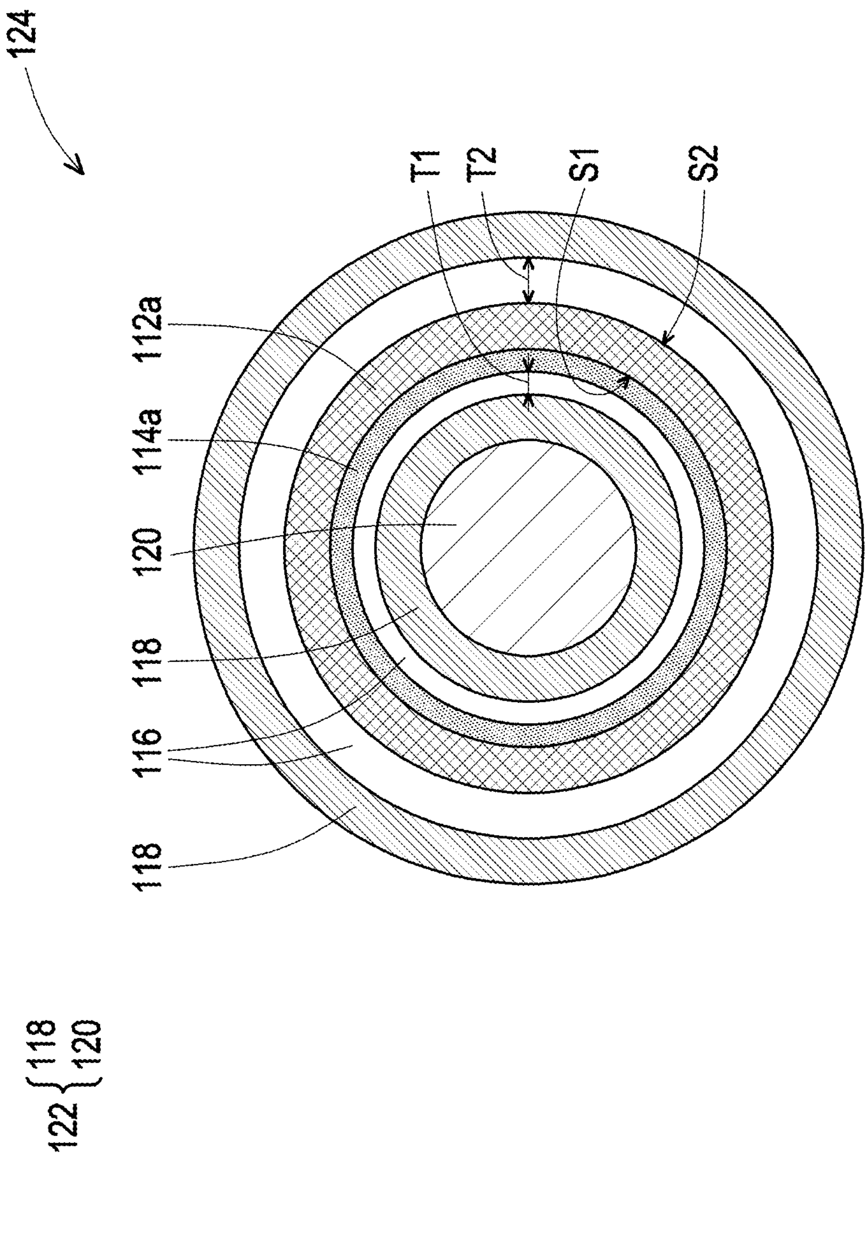


FIG. 2

SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Taiwan application serial no. 113108223, filed on Mar. 6, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to a semiconductor structure and a manufacturing method thereof, and more particularly, to a semiconductor structure including a capacitor and a manufacturing method thereof.

Description of Related Art

[0003] In a semiconductor structure (e.g., a dynamic random access memory (DRAM) structure) including a capacitor, when the capacitor bends or collapses, reliability of the semiconductor structure will be reduced. Therefore, how to improve the reliability of the semiconductor structure is a goal to work on.

SUMMARY

[0004] The disclosure provides a semiconductor structure and a manufacturing method thereof, which may improve reliability of the semiconductor structure.

[0005] The disclosure provides a semiconductor structure, including a substrate and a capacitor. The capacitor is located on the substrate. The capacitor includes a first electrode, a second electrode, a first dielectric layer, and a second dielectric layer. The first electrode is located on the substrate. The first electrode has a first surface and a second surface opposite to each other. The second electrode is located on the first electrode. The first dielectric layer is located between the first surface and the second electrode. The second dielectric layer is located between the first dielectric layer and the second electrode and between the second surface and the second electrode.

[0006] The disclosure provides a manufacturing method of a semiconductor structure, including the following. A substrate is provided. A capacitor is formed on the substrate. The capacitor includes a first electrode, a second electrode, a first dielectric layer, and a second dielectric layer. The first electrode is located on the substrate. The first electrode has a first surface and a second surface opposite to each other. The second electrode is located on the first electrode. The first dielectric layer is located between the first surface and the second electrode. The second dielectric layer is located between the first dielectric layer and the second electrode and between the second surface and the second electrode.

[0007] Based on the above, in the semiconductor structure and the manufacturing method thereof provided in the disclosure, the first dielectric layer is located between the first surface of the first electrode and the second electrode, and the second dielectric layer is located between the first dielectric layer and the second electrode and between the second surface of the first electrode and the second electrode. That is, the first dielectric layer and the second dielectric layer are provided on the first surface of the first electrode. In this way, the capacitor may be prevented from bending or collapsing, thereby improving the structural strength of the capacitor and the reliability of the semiconductor structure. In addition, the first dielectric layer may balance a thickness of capacitance dielectric layers (including the first dielectric layer and the second dielectric layer) located on the first surface (e.g., an inner surface) of the first electrode and the second surface (e.g., an outer surface) of the first electrode. In addition, the first dielectric layer may improve step coverage capability of the capacitance dielectric layers (including the first dielectric layer and the second dielectric layer) located on the first surface (e.g., the inner surface) of the first electrode.

[0008] In order for the aforementioned features and advantages of the disclosure to be more comprehensible, embodiments accompanied with drawings are described in detail below.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIGS. **1**A to **1**G are cross-sectional views of a manufacturing process of a semiconductor structure according to some embodiments of the disclosure.

[0010] FIG. **2** is a top view of a capacitor in FIG. **1**G.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

[0011] The embodiments are described in detail below with reference to the accompanying drawings, but the provided embodiments are not intended to limit the scope of the disclosure. For ease of understanding, the same elements in the following description will be denoted by the same reference numerals. In addition, the drawings are drawn only for the purpose of description, and are not drawn according to original sizes. Furthermore, the features in the top view and the features in the cross-sectional view are not drawn to the same scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0012] FIGS. **1**A to **1**G are cross-sectional views of a manufacturing process of a semiconductor structure according to some embodiments of the disclosure. FIG. **2** is a top view of a capacitor in FIG. **1**G. In addition, in FIG. **2**, some components in FIG. **1**G are omitted to clearly describe a configuration relationship between the components in FIG. **2**.

[0013] Referring to FIG. **1**A, a substrate **100** is provided. The substrate **100** may be a semiconductor substrate, such as a silicon substrate. In addition, although not shown in the drawing, there may be corresponding components on and/or in the substrate **100** depending on a type of the semiconductor structure. For example, the substrate **100** may have required components (not shown) such as isolation structures, doping regions, and/or buried word lines, and the substrate **100** may have required components (not shown) such as dielectric layers and/or interconnection structures (e.g., bit lines and contact windows). Here, descriptions thereof are omitted.

[0014] Next, a stack structure SS**1** may be formed on the substrate **100**. The stack structure SS**1** may include a dielectric layer **102**, a support layer **104**, a dielectric layer **106**, and a support layer **108**. The dielectric layer **102** is located on the substrate **100**. In some embodiments, a material of the dielectric layer **102** is, for example, oxides (e.g., silicon oxide). The support layer **104** is located on the dielectric layer **102**. In some embodiments, a material of the support layer **104** is, for example, nitrides (e.g., silicon nitride). The dielectric layer **106** is located on the support layer **104**. In some embodiments, a material of the dielectric layer **106** is, for example, oxides (e.g., silicon oxide). The support layer **108** is located on the dielectric layer **106**. In some embodiments, a material of the support layer **108** is, for example, nitrides (e.g., silicon nitride). In some embodiments, the stack structure SS1 may further include a stop layer **110**. The stop layer **110** is located between the dielectric layer **102** and the substrate **100**. In some embodiments, a material of the stop layer **110** is, for example, nitrides (e.g., silicon nitride).

[0015] Referring to FIG. **1**B, an opening OP**1** may be formed in the stack structure SS**1**. The opening OP**1** may penetrate the support layer **108**, the dielectric layer **106**, the support layer **104**, the dielectric layer **102**, and the stop layer **110**. In some embodiments, a patterning process may be performed on the stack structure SS**1** to form the opening OP**1** in the stack structure SS**1**. The patterning process may include a photolithography process and an etching process.

[0016] Referring to FIG. **1**C, an electrode material layer **112** may be formed conformally on the stack structure SS**1** and in the opening OP**1**. In some embodiments, a material of the electrode material layer **112** is, for example, titanium nitride. In some embodiments, a method of forming the electrode material layer **112** is, for example, chemical vapor deposition.

[0017] Referring to FIG. **1**D, a dielectric material layer **114** may be formed conformally on the electrode material layer **112**. In some embodiments, a material of the dielectric material layer **114** is, for example, a high dielectric constant material. In some embodiments, a material of the dielectric material layer **114** is, for example, hafnium oxide (HfO), zirconium oxide (ZrO), or niobium oxide (NbO). In some embodiments, a method of forming the dielectric material layer **114** is, for example, atomic layer deposition.

[0018] Referring to FIG. **1**E, the patterning process may be performed on the dielectric material layer **114** and the electrode material layer **112** to form a dielectric layer **114***a* and an electrode **112***a*. In some embodiments, the etching process (e.g., a dry etching process) may be performed on the dielectric material layer **114** and the electrode material layer **112** by using a patterned mask layer (e.g., a patterned photoresist layer and/or a patterned hard mask layer) (not shown) as a mask to form the dielectric layer **114***a* and the electrode **112***a*, and expose a portion of the support layer **108**.

[0019] In addition, a portion of the support layer **108** may be removed to expose a portion of the dielectric layer **106**. In some embodiments, a method of removing a portion of the support layer **108** is, for example, dry etching. Then, the dielectric layer **106** may be removed to expose the support layer **104**. In some embodiments, a method of removing the dielectric layer **106** is, for example, wet etching. Next, a portion of the support layer **104** may be removed to expose a portion of the dielectric layer **102**. In some embodiments, a method of removing a portion of the support layer **104** is, for example, dry etching. Then, the dielectric layer **102** may be removed. In some embodiments, a method of removing the dielectric layer **102** is, for example, wet etching.

[0020] Referring to FIG. **1**F, after the dielectric layer **102** is removed, a dielectric layer **116** may be formed conformally on the electrode **112***a*, the dielectric layer **114***a*, the support layer **104**, the support layer **108**, and the stop layer **110**. In some embodiments, a material of the dielectric layer **116** is, for example, the high dielectric constant material. In some embodiments, the material of the dielectric layer **116** is, for example, hafnium oxide, zirconium oxide, niobium oxide, or a composite material of zirconium oxide/aluminum oxide/zirconia (ZAZ). In some embodiments, a method of forming the dielectric layer **116** is, for example, atomic layer deposition or chemical vapor deposition.

[0021] Referring to FIG. **1**G, an electrode layer **118** may be formed conformally on the dielectric layer **116**. In some embodiments, a material of the electrode layer **118** is, for example, titanium nitride. In some embodiments, a method of forming the electrode layer **118** is, for example, chemical vapor deposition.

[0022] Then, an electrode layer **120** may be formed on the electrode layer **118**. In this way, an electrode **122** may be formed. In this embodiment, the electrode **122** may include the electrode layer **118** and the electrode layer **120**, but the disclosure is not limited thereto. The electrode layer **120** may be a single-layer structure or a multi-layer structure. In some embodiments, a material of the electrode layer **120** is, for example, doped silicon germanium (SiGe), tungsten, or a combination thereof. In some embodiments, a method of forming the electrode layer **120** is, for example, chemical vapor deposition.

[0023] Through the above method, a capacitor **124** may be formed on the substrate **100**. Hereinafter, a semiconductor structure **10** in the above embodiment will be described with reference to FIGS. **1**G and **2**. In addition, although a method of forming the semiconductor structure **10** is described by taking the above method as an example, the disclosure is not limited thereto.

[0024] Referring to FIGS. **1**G and **2**, the semiconductor structure **10** includes the substrate **100** and the capacitor **124**. In some embodiments, the semiconductor structure **10** may be a memory structure, such as a dynamic random access memory structure. In some embodiments, the capacitor **124** may be used as a capacitor in a dynamic random access memory.

[0025] The capacitor **124** is located on the substrate **100**. The capacitor **124** includes the electrode **112***a*, the electrode **122**, the dielectric layer **114***a*, and the dielectric layer **116**. The electrode **112***a* is located on the substrate **100**. In some embodiments, the electrode **112***a* may be used as a lower electrode of the capacitor **124**. The electrode **112***a* has a surfaces S1 and a surface S2 opposite to each other. In some embodiments, a cross-sectional shape of the electrode **112***a* may include a U-shape. In a case where the cross-sectional shape of the electrode **112***a* includes the U-shape, the surface S1 may be an inner surface of the electrode **112***a*, and the surface S2 may be an outer surface of the electrode **112***a*.

[0026] The electrode **122** is located on the electrode **112***a*. In some embodiments, the electrode **122** may be used as an upper electrode of the capacitor **124**. The electrode **122** may be a single-layer structure or a multi-layer structure. In this embodiment, the electrode **122** is a multi-layer structure as an example and includes the electrode layer **118** and the electrode layer **120**, but the disclosure is not limited thereto. The electrode layer **118** is located on the electrode **112***a*. The electrode layer **120** is located on the electrode layer **118**.

[0027] The dielectric layer **114***a* is located between the surface S1 and the electrode **122**. The dielectric layer **114***a* may prevent the capacitor **124** from bending or collapsing, thereby improving structural strength of the capacitor **124** and reliability of the semiconductor structure **10**. In some embodiments, the dielectric layer **114***a* may be in direct contact with the surface S**1**. A top-view shape of the dielectric layer **114***a* may be annular and may surround the electrode layer **120**. In some embodiments, the dielectric layer **114***a* and the electrode **112***a* may not include the same metal element. In some embodiments, a material of the dielectric layer **114***a* is, for example, the high dielectric constant material, thereby increasing capacitance of the capacitor **124**. In some embodiments, the material of the dielectric layer **114***a* is, for example, hafnium oxide, zirconium oxide, or niobium oxide. In some embodiments, the material of the dielectric layer **114***a* is, for example, a material with a dielectric constant greater than 20, such as hafnium oxide, zirconium oxide, niobium oxide, lanthanum oxide, or tantalum oxide. In some embodiments, the material of the dielectric layer **114***a* is, for example, a material with an energy gap greater than 8 electron volts, such as aluminum oxide.

[0028] The dielectric layer **116** is located between the dielectric layer **114***a* and the electrode **122** and between the surface S**2** and the electrode **122**. In some embodiments, the dielectric layer **114***a* may not be located between the surface S**2** and the dielectric layer **116**. In some embodiments, the dielectric layer **116** may be in direct contact with the surface S**2**. In some embodiments, the material of the dielectric layer **114***a* may be different from the material of the dielectric layer **116**. In other embodiments, the material of the dielectric layer **114***a* may be the same as the material of the dielectric layer **116**. In some embodiments, the material of the dielectric layer **116** is, for example, the high dielectric constant material, thereby increasing the capacitance of the capacitor **124**. In some embodiments, the material of the dielectric layer **116** is, for example, hafnium oxide, zirconium oxide, niobium oxide, or the composite material of zirconium oxide/aluminum oxide/zirconia (ZAZ). In some embodiments, the material of the dielectric layer **116** is, for example, the material with the dielectric constant greater than 20, such as hafnium oxide, zirconium oxide, niobium oxide, lanthanum oxide, or tantalum oxide. In some embodiments, the material of the dielectric layer **116** is, for example, the material with the energy gap greater than 8 electron volts, such as aluminum oxide.

[0029] In some embodiments, a thickness T**1** of the dielectric layer **116** located on the surface S**1** (e.g., the inner surface) may be less than a thickness T**2** of the dielectric layer **116** located on the surface S**2** (e.g., the outer surface). Therefore, a thickness of capacitance dielectric layers (including the dielectric layer **114***a* and the dielectric layer **116**) located on the surface S**1** (e.g., the inner surface) and the surface S**2** (e.g., the outer surface) may be balanced by the dielectric layer **114***a*.

[0030] The semiconductor structure **10** may further include the support layer **104** and the support layer **108**. The support layer **104** is located on the surface S**2** of the electrode **112***a*. The support layer **104** may be in direct contact with the electrode **112***a*. The support layer **108** is located on the surface S**2** of the electrode **112***a*. The support layer **108** may be in direct contact with the electrode **112***a*. The support layer **104** may be located between the support layer **108** and the substrate **100**. The dielectric layer **116** may be further located on the support layer **104** and the support layer **108**. The semiconductor structure **10** may further include the stop layer **110**. The stop layer **110** may be located between the support layer **104** and the substrate **100**. The dielectric layer **116** may be further located on the stop layer **110**.

[0031] In addition, details of each of the components in the semiconductor structure **10** (e.g., materials, configurations, formation methods, etc.) have been described in detail in the above embodiments, which will not be described in the following.

[0032] Based on the above embodiments, it may be seen that in the semiconductor structure **10** and the manufacturing method thereof, the dielectric layer **114***a* is located between the surface S**1** of the electrode **112***a* and the electrode **122**, and the dielectric layer **116** is located between the dielectric layer **114***a* and the electrode **122** and between the surface S**2** of the electrode **112***a* and the electrode **122**. That is, the dielectric layer **114***a* and the dielectric layer **116** are provided on the surface S**1** of the electrode **112***a*. In this way, the capacitor **124** may be prevented from bending or collapsing, thereby improving the structural strength of the capacitor **124** and the reliability of the semiconductor structure **10**. In addition, the dielectric layer **114***a* may balance the thickness of the capacitance dielectric layers (including the dielectric layer **114***a* and the dielectric layer **116**) located on the surface S**1** (e.g., the inner surface) of the electrode **112***a* and the surface S**2** (e.g., the outer surface) of the electrode **112***a*. In addition, the dielectric layer **114***a* may improve step coverage capability of the capacitance dielectric layers (including the dielectric layer **114***a* and the dielectric layer **116**) located on the surface S**1** (e.g., the inner surface) of the electrode **112***a*.

[0033] Although the disclosure has been described with reference to the above embodiments, they are not intended to limit the disclosure. It will be apparent to one of ordinary skill in the art that modifications to the described embodiments may be made without departing from the spirit and the scope of the disclosure. Accordingly, the scope of the disclosure will be defined by the attached claims and their equivalents and not by the above detailed descriptions.

What is claimed is:

1. A semiconductor structure, comprising:

a substrate; and

a capacitor located on the substrate and comprising:

- a first electrode located on the substrate and having a first surface and a second surface opposite to each other;
- a second electrode located on the first electrode;
- a first dielectric layer located between the first surface of the first electrode and the second electrode; and
- a second dielectric layer located between the first dielectric layer and the second electrode and between the second surface of the first electrode and the second electrode, wherein the second dielectric layer is located on opposite sides of the first electrode.

2. The semiconductor structure according to claim **1**, wherein the first dielectric layer is not located between the second surface of the first electrode and the second dielectric layer.

3. The semiconductor structure according to claim **1**, wherein a cross-sectional shape of the first electrode comprises a U-shape.

4. The semiconductor structure according to claim **1**, wherein the first dielectric layer and the first electrode do not include a same metal element.

5. The semiconductor structure according to claim **1**, wherein the first dielectric layer is in direct contact with the first surface of the first electrode.

6. The semiconductor structure according to claim **1**, wherein the second dielectric layer is in direct contact with the second surface of the first electrode.

7. The semiconductor structure according to claim **1**, wherein a material of the first dielectric layer comprises a high dielectric constant material.

8. The semiconductor structure according to claim **1**, wherein a material of the first dielectric layer comprises hafnium oxide, zirconium oxide, niobium oxide, lanthanum oxide, or tantalum oxide.

9. The semiconductor structure according to claim **1**, wherein a thickness of the second dielectric layer located on the first surface of the first electrode is less than a thickness of the second dielectric layer located on the second surface of the first electrode.

10. The semiconductor structure according to claim **1**, wherein a material of the first dielectric layer comprises a material with a dielectric constant greater than 20.

11. The semiconductor structure according to claim **1**, wherein a material of the first dielectric layer comprises a material with an energy gap greater than 8 electron volts.

12. The semiconductor structure according to claim **1**, further comprising:

- a first support layer located on the second surface of the first electrode; and
- a second support layer located on the second surface of the first electrode, wherein the first support layer is located between the second support layer and the substrate.

13. The semiconductor structure according to claim **12**, wherein the second dielectric layer is further located on the first support layer and the second support layer.

14. A manufacturing method of a semiconductor structure, comprising:

- providing a substrate; and
- forming a capacitor on the substrate, wherein the capacitor comprises:
  - a first electrode located on the substrate and having a first surface and a second surface opposite to each other;
  - a second electrode located on the first electrode;
  - a first dielectric layer located between the first surface and the second electrode; and
  - a second dielectric layer located between the first dielectric layer and the second electrode and between the second surface and the second electrode, wherein the second dielectric layer is located on opposite sides of the first electrode.

15. The manufacturing method of the semiconductor structure according to claim **14**, wherein a method of forming the first electrode and the first dielectric layer comprises:

- forming a stack structure on the substrate;
- forming an opening in the stack structure;
- forming an electrode material layer conformally on the stack structure and in the opening;
- forming a dielectric material layer conformally on the electrode material layer; and
- performing a patterning process on the dielectric material layer and the electrode material layer to form the first dielectric layer and the first electrode.

16. The manufacturing method of the semiconductor structure according to claim **15**, wherein a method of forming the dielectric material layer comprises atomic layer deposition.

17. The manufacturing method of the semiconductor structure according to claim **15**, wherein the stack structure comprises:

- a third dielectric layer located on the substrate;
- a first support layer located on the third dielectric layer;
- a fourth dielectric layer located on the first support layer; and
- a second support layer located on the fourth dielectric layer.

18. The manufacturing method of the semiconductor structure according to claim **17**, further comprising:

- removing a portion of the second support layer to expose a portion of the fourth dielectric layer;
- removing the fourth dielectric layer to expose the first support layer;
- removing a portion of the first support layer to expose a portion of the third dielectric layer; and
- removing the third dielectric layer.

19. The manufacturing method of the semiconductor structure according to claim **18**, wherein a method of forming the second dielectric layer comprises:

- after removing the third dielectric layer, forming the second dielectric layer conformally on the first electrode, the first dielectric layer, the first support layer, and the second support layer.

20. The manufacturing method of the semiconductor structure according to claim **14**, wherein a method of forming the second dielectric layer comprises atomic layer deposition or chemical vapor deposition.

* * * * *